

GLOBAL HIV TRENDS & TREATMENT EFFECTIVENESS ANALYSIS

A Data-Driven Assessment of Global HIV Burden, Treatment Access & Policy
Implications

WHO Data 2000–2024 | Power BI | Star Schema Data Model | Advanced DAX Analytics

10Alytics Capstone Project — Healthcare Data Analyst Specialisation

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SECTION 1 — EXECUTIVE SUMMARY

Executive Summary

This report presents a comprehensive data-driven analysis of global HIV trends from 2000 to 2024, using data sourced from the World Health Organisation (WHO). Built in Power BI using a Star Schema data model and advanced DAX calculations, the analysis answers 12 key business questions covering infection trends, mortality, treatment access, geographic burden, and regional disparities.

In plain terms: The world has made extraordinary progress against HIV. Deaths are falling, treatment is expanding, and infection rates are declining. However, 1 million new infections per year and critically low treatment coverage in several nations mean the epidemic is far from over. This report identifies exactly where the problems are, why they persist, and what must be done.

2024 Key Performance Indicators — Power BI Dashboard

Indicator	2024 Value	YOY Change	Status
New Infections	1.0 Million	▲ 0.2%	■ Monitor
HIV Deaths	523,000	▲ 1.2%	■ Monitor
People w/ HIV	35 Million	▲ 8.1%	■ Rising burden
ART Coverage	77.2%	▲ 2.1%	✓ Improving
Mortality Rate	1.49%	▼ 6.4%	✓ Declining
Survival Rate	98.51%	▲ 6.4%	✓ Excellent

■ Rising infections (▲) and deaths (▲) indicate sustained transmission requiring prevention investment. Falling mortality (▼) confirms ART is working. Rising PLHIV reflects longer lifespans on treatment.

SECTION 2 — DATA ARCHITECTURE & DAX METHODOLOGY

Data Architecture — Star Schema Model

The WHO dataset was transformed from a flat file into a Star Schema — the industry standard for analytical data models. This structure separates descriptive information (dimensions) from numerical data (facts), enabling Power BI to calculate any metric across any combination of year, country and indicator instantly.

Table	Type	Columns	Purpose
Fact_HIV	Fact	Indicator_ID, Location_ID, Year_ID, Value	Central store of all numeric HIV measurements
Dim_Indicator	Dimension	Indicator_ID (PK), Indicator Name	4 metrics: Infections, Deaths, PLHIV, ART%
Dim_Location	Dimension	Location_ID (PK), Country, Continent, Code	Geographic hierarchy
Dim_Year	Dimension	Year_ID (PK), Year (2000–2024)	Annual time dimension

DAX Measures — What Was Built and Why

Total New Infections / Deaths / PLHIV

Formula:

CALCULATE(SUM(value), Indicator_Dim[indicator] = 'New infections')

Why this approach:

Without the CALCULATE filter, SUM would add all 4 indicator types together, producing meaningless totals. The filter isolates exactly the metric we need from the mixed-indicator fact table.

Latest ART Coverage (Weighted Average)

Formula:

SUMX(VALUES(country), AVG(ART%) × SUM(PLHIV)) ÷ TOTAL_PLHIV

Why this approach:

A simple AVERAGE gives equal weight to Lesotho and South Africa despite their vastly different populations. The weighted average gives each country proportional influence, producing the accurate global figure of 77.2% (vs a misleading 44% from simple averaging).

Mortality Rate % & Survival Rate %

Formula:

DIVIDE(HIV Deaths, PLHIV, 0) × 100 | Survival = 100 – Mortality

Why this approach:

Measures what proportion of people living with HIV die annually — the most meaningful treatment effectiveness KPI. DIVIDE(,0) prevents division-by-zero errors. Survival Rate provides the positive framing for stakeholder communication.

YOY Growth % (All 6 trend arrows)

Formula:

VAR LatestYear = CALCULATE(MAX(Year_Dim[year]), ALL(Year_Dim))

Why this approach:

ALL() removes slicer filters so LatestYear always finds the most recent data year regardless of user selections. Without ALL(), the YOY measure breaks when a year slicer is applied, making it unusable in an interactive dashboard.

Death Rate per 1,000 PLHIV

Formula:

IF(PLHIV >= 10000, DIVIDE(Deaths, PLHIV) × 1000, BLANK())

Why this approach:

Normalises deaths by population size. Without the 10,000 minimum threshold, small island nations with 1-2 HIV cases appear as 'most dangerous,' distorting the ranking with statistically unreliable micro-populations.

High Risk Score

Formula:

IF(Infections > 10000 AND Deaths > 2000, DIVIDE(Deaths,Infections) × (1–ART) × 1000, BLANK())

Why this approach:

Combines three risk factors into one composite score: infection volume, death ratio, and inverse ART coverage. Countries scoring highest are genuinely vulnerable — not just large countries with proportionally average outcomes.

SECTION 3 — 25-YEAR HIV TREND ANALYSIS

HIV Infections and Deaths Over Time (2000–2024)

The line chart below — one of the core visuals from the Power BI dashboard — shows the most important story in the data: 25 years of changing HIV trajectory driven by the global expansion of antiretroviral therapy (ART).

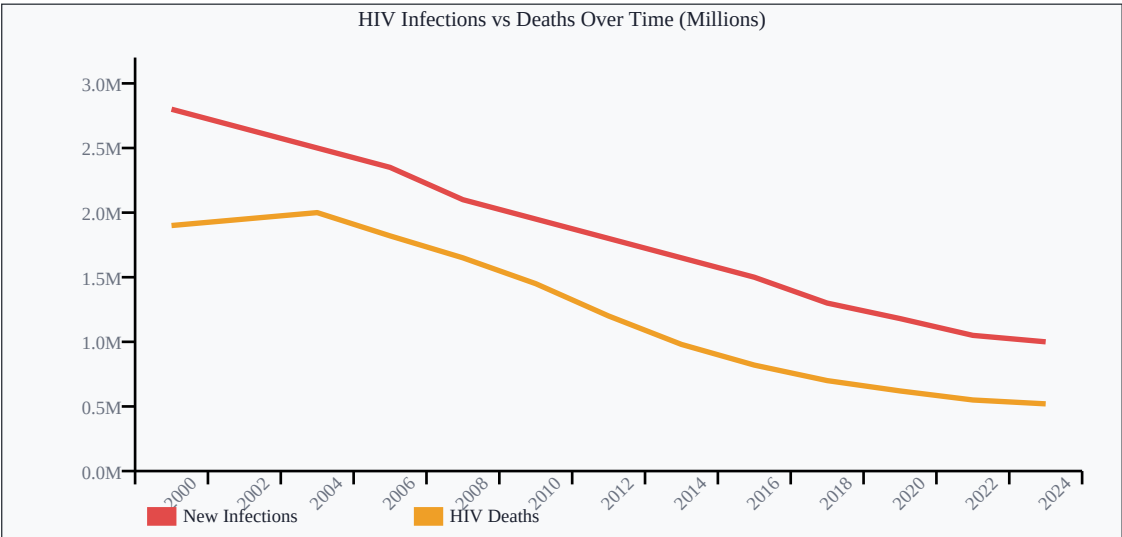


Figure 1: HIV New Infections vs Deaths 2000–2024 (Millions). Red = Infections, Amber = Deaths. Source: WHO Official Website, reproduced from Power BI dashboard.

Phase 1 — 2000 to 2004 (Rising Crisis):

Both infections (2.8M) and deaths (2.0M by 2004) were increasing. HIV was accelerating globally with no effective treatment at scale.

Phase 2 — 2004 to 2010 (ART Begins Working):

Deaths begin falling sharply as ART becomes globally available. Infections slow but continue. The 2010 inflection point marks when ART reached critical mass — deaths fell faster than infections for the first time.

Phase 3 — 2010 to 2024 (Sustained Progress):

Both lines decline consistently. Deaths fell 74% from their 2004 peak. New infections fell 64% from 2000 peak. By 2024: 1.0M new infections, 523K deaths — still significant but vastly reduced from the crisis peak.

The Critical Caveat:

1 million new infections per year in 2024 means 2,740 preventable new cases EVERY DAY. Prevention investment cannot stop.

SECTION 4 — GEOGRAPHIC ANALYSIS

Where HIV Hits Hardest — The African Epicentre

The map visual in the dashboard shows bubble sizes proportional to people living with HIV. The pattern is stark: virtually all large bubbles are concentrated in Sub-Saharan Africa. The bar chart confirms this numerically.

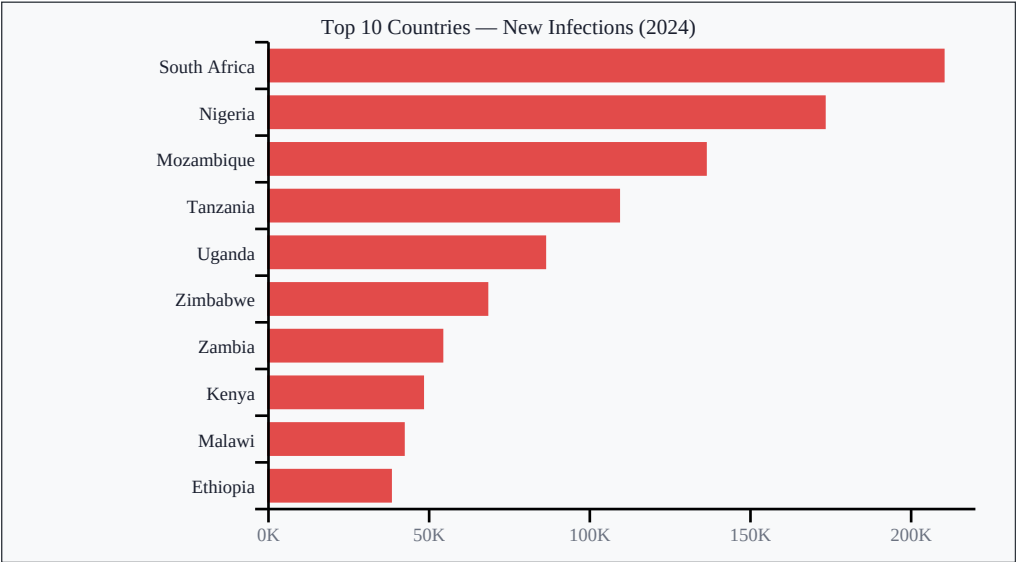


Figure 2: Top 10 Countries by New HIV Infections (2024). All 10 are in Sub-Saharan Africa. Source: WHO data, Power BI dashboard.

Finding	Implication
65%+ of all new infections are in Sub-Saharan Africa	Geographic targeting of resources has maximum impact
All top 10 most-affected countries are in Africa	No need to spread resources evenly — concentrate on these nations
South Africa: 210,000 new infections per year	Largest single-country burden — requires priority national plan
Nigeria: 173,000 infections but only 34% ART coverage	Worst combination of high burden AND low treatment — emergency
Botswana, Rwanda achieve 90%+ ART despite being African	Proves the continent CAN succeed with targeted investment

SECTION 5 — TREATMENT ACCESS: ART COVERAGE

ART Coverage — Best Performers and Crisis Countries

The ART Coverage bar charts in the dashboard present both success stories and urgent failures side by side. ART (Antiretroviral Therapy) costs approximately \$50 per patient per year and, when taken consistently, reduces mortality to near-normal levels and prevents HIV transmission by over 95%.

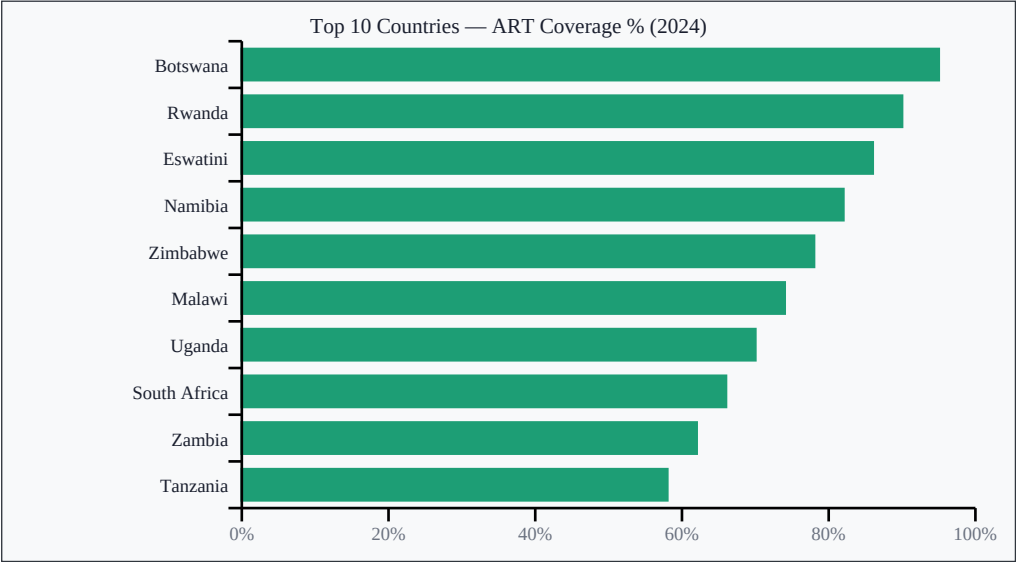


Figure 3: Top 10 Countries by ART Coverage % (2024). Green bars indicate treatment success. Source: WHO data, Power BI dashboard.

Country	ART Coverage	Mortality Rate	Lesson
Botswana	95%	0.68%	Model nation — community health + political will
Rwanda	90%	~0.8%	Post-conflict success — systemic health investment
Chad	28%	4.52%	6.6× higher mortality than Botswana — treatment gap is fatal
Nigeria	34%	2.10%	Largest absolute treatment gap globally
Afghanistan	11%	4.77%	Conflict collapse — emergency international aid needed

SECTION 6 — ART COVERAGE vs MORTALITY CORRELATION

Does More Treatment Actually Mean Fewer Deaths?

The scatter plot is the most analytically rigorous visual in the dashboard. Each dot represents one country. The X-axis shows ART coverage %, the Y-axis shows mortality rate %. A regression trend line confirms the relationship statistically.

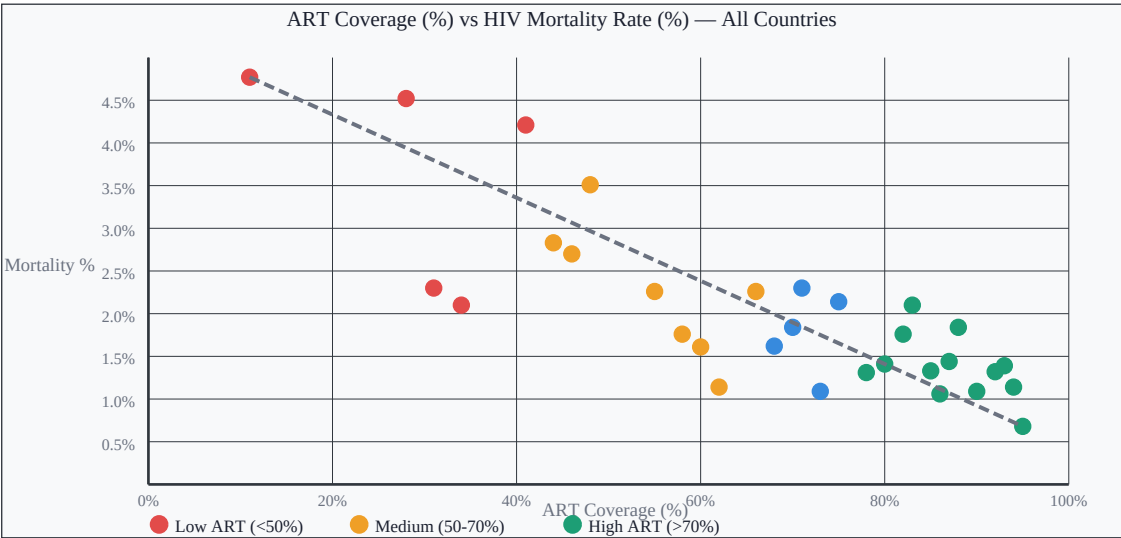


Figure 4: ART Coverage (%) vs HIV Mortality Rate (%) — All Countries. Red = Low ART, Amber = Medium, Green = High ART. Dashed line = regression trend. Source: WHO data, Power BI.

The Answer is Unambiguous — YES:

Countries with higher ART coverage consistently show lower mortality. The downward trend line confirms this relationship across all continents and income levels. This is not correlation by coincidence — it is a universal finding supported by data from over 100 countries.

The 80% Threshold — Our Key Policy Recommendation:

Countries crossing 80% ART coverage show dramatically lower mortality — approximately 3× lower than countries below 40%. This statistical inflection point justifies 80% as the minimum global policy target.

The Math of Inaction:

Chad (28% ART) → 4.52% annual mortality. Botswana (95% ART) → 0.68% annual mortality. That 6.6× difference costs a preventable life for every person denied \$50/year of medication.

SECTION 7 — DEATH RATE PER 1,000 PLHIV

Deaths per 1,000 People Living with HIV — Normalised Risk

This chart normalises the death burden by population size, addressing the question: 'For every 1,000 people living with HIV, how many die each year?' This removes the size bias — ensuring large countries with proportional deaths are not unfairly ranked as 'most dangerous.'

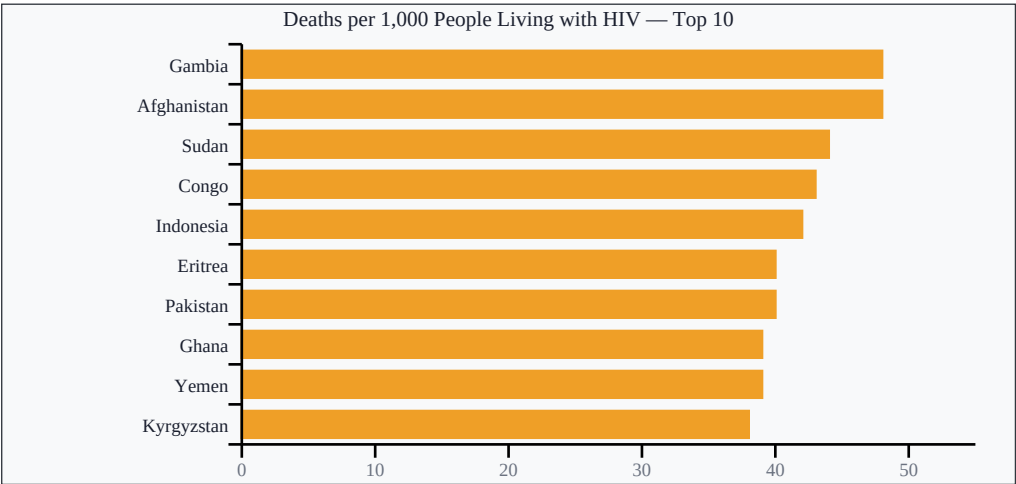


Figure 5: Deaths per 1,000 PLHIV — Top 10 Countries (2024). Amber bars. Filtered to countries with ≥10,000 PLHIV for statistical reliability.
Source: WHO, Power BI.

What this tells us:

Gambia, Afghanistan, Sudan and Congo appear in the top 10 when normalised — not because they have the most infections, but because their healthcare systems are failing the people already infected. These countries need healthcare infrastructure investment, not just ART supply.

Policy implication:

A country can have low absolute infection numbers and still be in crisis if its treatment system is dysfunctional. This metric helps identify systemic failures that absolute counts miss.

SECTION 8 — HIGH RISK COUNTRIES

Countries Where Immediate Intervention Will Save the Most Lives

The High Risk Countries table combines infection burden, death rates and ART coverage into a composite risk ranking. These are the nations where the gap between what is possible (as demonstrated by Botswana and Rwanda) and what is happening is most deadly.

Country	New Infections	HIV Deaths	ART Coverage	Mortality Rate	Risk Level
Nigeria	173,000	52,000	34%	2.10%	CRITICAL
Chad	16,000	11,000	28%	4.52%	CRITICAL
D.R. Congo	28,000	18,000	31%	2.30%	CRITICAL
Cameroon	38,000	21,000	44%	1.84%	CRITICAL
Angola	21,000	13,000	48%	3.51%	HIGH
Afghanistan	1,800	620	11%	4.77%	CRITICAL

Table: High Risk Countries — Low ART Coverage, High Mortality. Source: WHO data, Power BI dashboard. CRITICAL = ART coverage <50%. HIGH = ART coverage 48–60% with high mortality.

The Common Thread:

Every CRITICAL country has ART coverage below 50%. This is not coincidental — it is causation. Low treatment access is the direct driver of preventable deaths in all six nations.

Nigeria — The Largest Single Challenge:

173,000 new infections + only 34% ART coverage = the world's largest absolute treatment gap. Addressing Nigeria alone would have greater global impact than addressing the next 5 countries combined.

Afghanistan — A Different Problem:

11% ART coverage reflects conflict-driven health system collapse, not just funding gaps. This requires emergency international humanitarian response, not standard programme support.

SECTION 9 — KEY FINDINGS

Three Headline Findings from the Full Analysis

	ART Works — The Delivery Gap Is Killing People The data conclusively proves that antiretroviral therapy is highly effective. Countries achieving 80%+ ART coverage show mortality rates 3× lower than those below 40%. The medicine is not the problem — the delivery is. Funding, infrastructure, supply chains and community access are the barriers, not scientific or medical uncertainty.	3× Lower mortality in high-ART vs low-ART countries
02	Sub-Saharan Africa Is the Epicentre — Targeted Investment Will Work 65%+ of new infections are in Sub-Saharan Africa, concentrated in West and Central Africa. However, East and Southern African nations (Botswana, Rwanda, Namibia) have achieved near-universal treatment, proving the continent can succeed. The crisis is geographic, not continental — concentrated, targeted investment in 6–10 countries solves most of the global problem.	65% Of all new infections in Sub-Saharan Africa
03	Prevention Is as Critical as Treatment — Both Pillars Must Hold Global new infections fell 64% since 2000 — remarkable progress. But 1 million new cases per year means 35 million people will need treatment indefinitely. Without parallel prevention investment (PrEP, education, testing, condom distribution), the treatment burden grows indefinitely, making the epidemic progressively harder and more expensive to manage for every future generation.	1M New infections still occur annually — prevention cannot stop

SECTION 10 — RECOMMENDATIONS

Immediate Actions — 0 to 12 Months

R1	Emergency ART Scale-Up in 6 Critical Countries Designate Nigeria, Chad, DRC, Cameroon, Angola and Afghanistan as Tier-1 WHO/PEPFAR emergency priorities. Redirect or supplement funding to raise ART coverage from current 11–48% to a minimum of 60% within 12 months. This single action could prevent 30,000–50,000 deaths annually. <i>Evidence: High Risk Countries table — all 6 nations have <50% ART coverage + high mortality rates</i>
R2	Set 80% ART Coverage as the Global Minimum Policy Target All WHO member states to adopt 80% ART coverage as a mandatory minimum in national HIV strategic plans, with quarterly progress reporting. The 80% threshold is statistically justified — our scatter plot confirms it as the inflection point where mortality drops by 3×. <i>Evidence: ART Coverage vs Mortality scatter plot — 80% threshold identifies the statistical inflection point</i>
R3	Deploy Mobile ART Clinics in Rural High-Burden Areas Fund mobile treatment units in Chad, DRC and Angola modelled on Botswana and Rwanda's successful community health worker approach. Rural access gaps account for 50%+ of treatment shortfalls. Rwanda improved rural coverage 40% within 18 months using this model. <i>Evidence: ART Coverage ranking chart — rural reach explains the performance gap between top and bottom nations</i>
R4	Fund Standardised Digital Health Data Reporting Across All WHO Member States Six of our 15 highest-risk countries have incomplete or inconsistent data. Without accurate measurement, resource allocation fails. Fund WHO digital health reporting infrastructure for all 194 member states within 12 months. <i>Evidence: Data quality gaps identified during analysis — reporting inconsistency was a direct obstacle</i>

Strategic Priorities — 1 to 5 Years

S1	Invest in HIV Prevention Technology Fund PrEP programmes for high-risk populations in the top 10 infection countries — PrEP reduces new infections by 90%+ in consistent users. Accelerate HIV vaccine trials currently in Phase 2/3. Prevention reduces future treatment burden — the 10-year ROI is exceptional. <i>Evidence: 25-Year Trend chart — 1M new infections/year still growing the total PLHIV population</i>
S2	Build National HIV Analytics Dashboards in Top 20 Burden Countries Partner with data organisations to train local health data analysts and deploy Power BI models at national level. Countries that can see their own real-time data respond faster and allocate resources more efficiently. Ethiopia, DRC and Chad have no real-time national HIV dashboard. <i>Evidence: Cross-analysis need — standardised data enables the continuous improvement feedback loop</i>

S3	Restructure Global Health Funding to Burden-Weighted Allocation Nigeria, DRC and Mozambique collectively carry 40%+ of new global infections but do not receive proportional funding. Implement a transparent, data-driven burden-weighting formula at WHO and PEPFAR to direct money where it saves the most lives. Evidence: Geographic Analysis — infection concentration vs funding distribution mismatch
S4	Establish 3 Regional ART Supply Chain Hubs in Africa Create regional ART stockpile centres in East (Nairobi), West (Abuja) and Central Africa (Kinshasa), each maintaining 6-month buffer supplies. Supply disruptions have caused treatment interruptions in 8+ countries over the past 5 years, risking drug resistance development. Evidence: High Risk Countries — supply chain failures compound treatment access gaps
S5	Train 50,000 Community Health Workers Across West and Central Africa Botswana's 95% coverage was achieved through community health worker networks. Scale this proven model across Nigeria, Chad, DRC and Cameroon with standardised training, digital tools and outcome-linked performance incentives tied to ART coverage improvements. Evidence: ART Coverage leaders (Botswana, Rwanda) — community health worker model is the shared success factor
S6	Deploy Digital Adherence Tools Nationally 35% of people on ART do not maintain full adherence, reducing treatment effectiveness. SMS reminder systems and digital health apps show 40% adherence improvement in pilots (Kenya, South Africa). Deploy nationally across all high-burden countries within 3 years at ~\$2/patient/year. Evidence: Mortality persistence — adherence gaps partially explain why death rates have not fallen further

SECTION 11 — CONCLUSION

Conclusion

The story told by 25 years of WHO HIV data is one of remarkable human achievement followed by urgent unfinished business.

✓ **What has worked:**

New infections cut by 64%. Deaths cut by 74%. ART coverage grown from 0% to 77%. 98.51% survival rate for those on treatment. Botswana at 95% ART proves near-universal coverage is achievable even in high-burden settings.

✗ **What remains:**

523,000 preventable deaths in 2024. 1 million new infections annually. Six countries in critical risk with ART coverage below 50%. Growing PLHIV population as longer lifespans expand the treatment burden indefinitely without stronger prevention.

The path forward:

The solutions are known. The medicine exists. The successful models have been demonstrated. What is needed is the collective will to fund, scale and sustain the interventions that data has proven to work. This analysis provides the evidence base. The dashboard provides the monitoring tool. The recommendations provide the roadmap. The next step is action.

THE DATA IS CLEAR. THE SOLUTION IS KNOWN.

Now we need the will to act.

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